CS Executive Sarv (New Syllabus) June 2025 (Module 1)

Financial Management

DPP: 2

Cost of Capital

Q1 Which of the following cost of capital require tax adjustment?
(A) Cost of Equity
(B) Cost of Debentures
(C) Cost of Retained Earning
(D) Cost of Preference Share Capital

Q2 Preference dividend is paid on:
(A) Redeemable value
(B) Face value
(C) Issue price
(D) Market value

Q3 Cost of Preference Share Capital is determined by the:
(A) Different Rate of Dividend
(B) Fixed Rate of Dividend
(C) Increasing Rate of Dividend
(D) Decreasing Rate of Dividend

Q4 Cost of capital refers to?
(A) Flotation cost
(B) Dividend
(C) Required rate of return
(D) None of the above

Q5 Which of the following sources of funds has an implicit cost of capital?
(A) Equity share capital
(B) Debentures
(C) Retained earning
(D) Preference Share

Q6 CD Ltd. issued 10,000, 14% debentures of Rs. 100 each at a discount of 10%. The debentures are irredeemable. Cost of issue is 2% and the rate of tax is 35%. Calculate cost of debt.
(i) Before tax
(ii) After tax

Q7 XYZ Ltd. issued 10,000, 20% debentures of Rs. 100 each at a discount of 8%. The debentures are irredeemable. Cost of issue is 2% and the rate of tax is 50%. Calculate cost of debt.
(i) Before tax
(ii) After tax

Q8 RK Ltd. is willing to issue 5,000 7% Debentures of Rs. 100 each and for which the company will have to incur the following expenses:
Underwriting commission 1%, Brokerage 0.50%, Printing and Other Expenses Rs. 2500. Assuming tax rate at 50% find out the cost of debt capital.
(i) Before tax
(ii) After tax

Q9 AB Company issued 1,000 ten-years 8% Debentures of Rs. 100 each at 4% discount. These debentures are to be redeemed after 10 years at 5% premium. The cost of issue is 2%. Assuming tax rate at 50%, Calculate the cost of debt capital.
(i) Before tax
(ii) After tax

Q10 XYZ issued 1,000 five-years 10% Debentures of Rs. 100 each. What is the cost of debt if the firm pays 35% tax?

case(a) Redeemed at 5% premium

case(b) Redeemed at 5% discount

Answer Key

Q1	(B)
Q2	(B)
Q3	(B)
Q4	(C)
Q5	(C)
Q6	(i) 0.1590 or 15.90% (ii) 0.1034 or 10.34%
Q7	(i) 0.2222 or 22.22% (ii) 0.1111 or 11.11%
Q8	(i) 0.0714 or 7.14% (ii) 0.0357 or 3.57%
Q9	(i) 0.0914 or 9.14% (ii) 0.0512 or 5.12%
Q10	case(a) 0.0731 or 7.31% case(b) 0.0564 or 5.64%

Hints & Solutions

Q1 Text Solution:

Cost of Debentures require Tax adjusment.

Q2 Text Solution:

Preference Dividend is paid on face value

Q3 Text Solution:

Cost of Preference Share is calculated on Fixed Rate Of Dividend.

Q4 Text Solution:

Cost of capital refers to require rate of return.

Q5 Text Solution:

Retained earning is a source of fund which has an implicit cost of capital..

Q6 Text Solution:

Here,

Interest = 140,000

Tax = 35%

Net Proceeds = (10,000 debentures x100)- (10% discount on face value) – (2% cost of issue on face value)

= 10,00,000- (10% of 10,00,000)- (2% of 10,00,000)

= 10,00,000- 100,000- 20,000

= 8,80,000

Kd(before tax) = $\frac{I}{NP}$

= $\frac{1,40,000}{8,80,000}$

= 0.1590 or 15.90%

Kd (after tax) = $\frac{I(1-t)}{NP}$

= $\frac{1,40,000(1-0.35)}{8,80,000}$

= 0.1034 or 10.34%

Q7 Text Solution:

Here,

Interest= 200,000

Tax= 50%

Net Proceed = (10,000 debentures x100)- (8% discount on face value) – (2% cost of issue on face value)

= 10,00,000- (8% of 10,00,000)- (2% of 10,00,000)

= 10,00,000- 80,000- 20,000= 900,000

Kd (before tax) = $\frac{I}{NP}$

= $\frac{2,00,000}{9,00,000}$

= 0.2222 or 22.22%

Kd (after tax) = $\frac{I(1-t)}{NP}$

= $\frac{2,00,000(1-0.50)}{9,00,000}$

= 0.1111 or 11.11%

Q8 Text Solution:

Here,

Interest= 35,000

Tax= 50%

Net Proceed= (5,000 debentures x100)- (1%

Underwriting commission on face value) –(0.05% Brokerage on face value) – (Rs2500 Other exepense)

= 500,000- (1% of 5,00,000)- (0.05% of 5,00,000)- (2500)

= 500,000- 5000- 2500- 2500

= 4,90,000

Kd (before tax) = $\frac{I}{NP}$

= $\frac{35,000}{4,90,000}$

= 0.0714 or 7.14%

Kd (after tax) = $\frac{I(1-t)}{NP}$

= $\frac{35,000(1-0.50)}{4,90,000}$

= 0.0357 or 3.57%

Q9 Text Solution:

Here,

I = 8% of 1,00,000 = Rs. 8,000

Tax = 50%

NP = (1000 debentures x100)- (4% Discount on face value)- (2% cost of issue)

= 1,00,000- (4% of 1,00,000)- (2% of 1,00,000)

=1,00,000 – 4,000 – 2,000

= Rs. 94,000

RV = (1000 debentures x100)+ (5% premium on face value)

= 1,00,000 + (5% of 1,00,000)

= 1,00,000 + 5,000

= Rs. 1,05,000

Kd (before tax) = $\frac{I+\frac{(RV-NP)}{n}}{\frac{RV+NP}{2}}$

$= \frac{8000+\frac{(1,05,000-94,000)}{10}}{\frac{1,05,000+94,000}{2}}$

= 0.0914 or 9.14%

Kd (after tax) = $\frac{I\left(1-t\right)+\frac{(RV-NP)}{n}}{\frac{RV+NP}{2}}$

$= \frac{8000\left(1-0.050\right)+\frac{(1,05,000-94,000)}{10}}{\frac{1,05,000+94,000}{2}}$

= 0.0512 or 5.12%

Q10 Text Solution:

Here,

Interest= 10,000

Net Proceed= (1000 debentures x 100)= 100,000

Redeemable Value= 100,000 + (5% Premium on face value)

= 100,000+ 5000

= 1,05,000

Case(a): Kd= $\frac{I\left(1-t\right)+\frac{(RV-NP)}{n}}{\frac{RV+NP}{2}}$

$= \frac{10,000\left(1-0.35\right)+\frac{(1,05,000-1,00,00)}{5}}{\frac{1,05,000+1,00,000}{2}}$

$= \frac{7500}{1,02,500}$

= 0.0731 or 7.31%

case(b): Here,

Interest= 10,000

Tax = 35%

Net Proceed= (1000 debentures x 100) = 100,000

Redeemable value= 100,000 - (5% Discount on face value)

= 100,000- 5000

= 95,000

Case(b): Kd= $\frac{I\left(1-t\right)+\frac{(RV-NP)}{n}}{\frac{RV+NP}{2}}$

$= \frac{10,000\left(1-0.35\right)+\frac{(95,000-1,00,000)}{5}}{\frac{95,000+1,00,000}{2}}$

$= \frac{5500}{97500}$

= 0.0564 or 5.64%